ConsensusFinality

Commit the approved result. Verify its actual on-chain effect.

ELI16 · THE IDEA IN PLAIN LANGUAGE

A signed sealed envelope: retry the same envelope after a lost reply, then read the ledger to check what was actually stored.

Why it matters: Submission, inclusion and finalized effect are different observations. Exact approval and durable signed intent prevent retries from changing the authorized artifact.

Through M5 · 17 October 2026 · 12–15 minute talk · evidence and proposals kept separate

One module in a system of fifteen

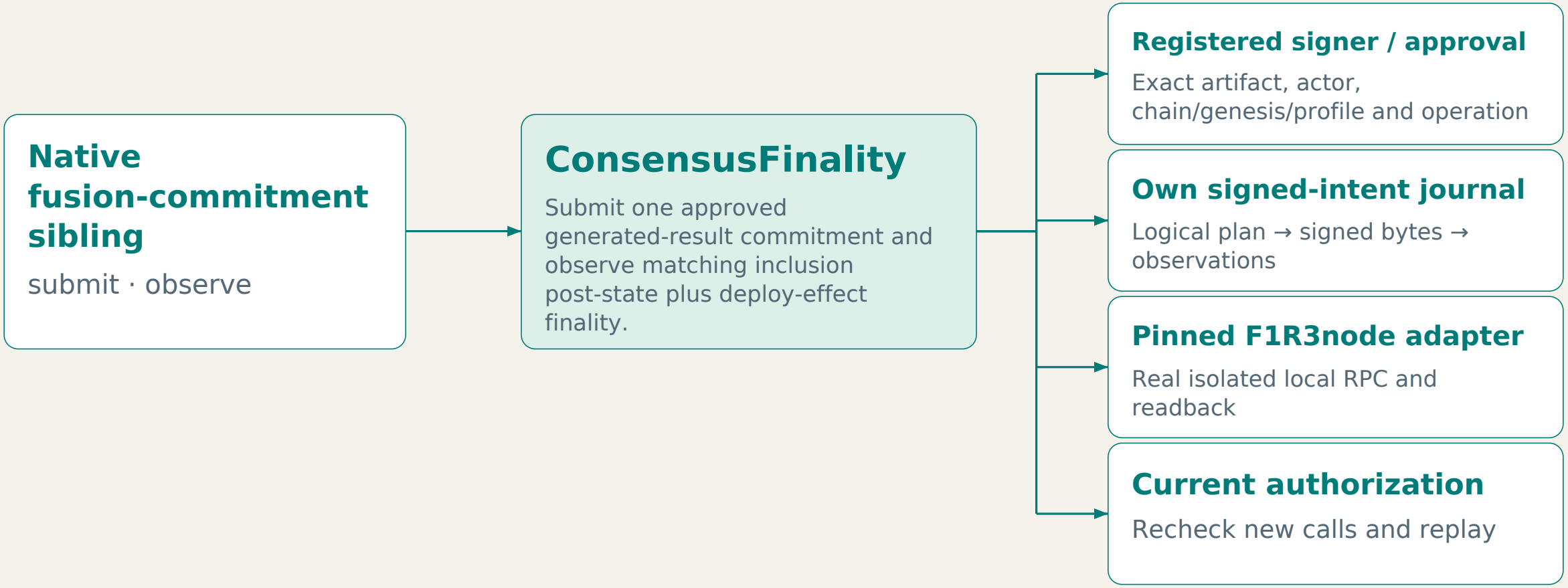
The highlighted responsibility is the subject of this talk; placement and process count are separate decisions.

Persistence	Knowledge access	Cognition	Coordination + policy
DA storage	Relations	Episodes · C1	Messaging
AtomStorage · C0	Search projection	Beliefs · C2	Execution
ConsensusFinality		PLNKernel	Registry
		Attention · C3	Resolver
		Inference · C4	Authorization

Submit one approved generated-result commitment and observe matching inclusion post-state plus deploy-effect finality.

Dependencies explain the boundary

Arrows point from caller to dependency. Stores and runtime internals are labelled explicitly.



One responsibility does not imply one process. A dependency does not grant access to its private tables.

Responsibility, state and scope

The owner is authoritative for its own state; other responsibilities are deliberately separate.

OWNS / GUARANTEES

Logical and signed intents

Stable timestamp, deployment identity and full signed request

Verified observations

Exact approved tuple, inclusion block/post-state and matching deploy effect

OUTSIDE THIS RESPONSIBILITY

Consensus implementation

No rewritten consensus, public network or new M5 chain operation

Cognitive correctness

Finality does not prove reasoning, belief truth or off-chain retention

f1r3-generated-post-state/2 · clean F1R3node 9ec1096 · fresh journal

The executable surface

These operations and implementation seams define the supported scope; versions and profiles remain explicit.

OPERATION / SEAM	RESPONSIBILITY	CONTRACT CONSEQUENCE
submit	Validate approval; persist plan before signing, bytes before send	Replay reuses the exact signed deployment
observe	Check inclusion, post-state value and deploy-effect finality	Finalized block alone is insufficient
Independent readback	Read same effect with read-only access	Separate process; no caller journal or signing key

One local validator; bounded call-driven observation, no public-network attachment.

A worked example through the boundary

M4 REAL isolated F1R3node · generated workflow, test genesis/key and fixture counterparty

01 · STEP

Accept result

Real Hive inspect finishes; retrieve approved bytes.

02 · STEP

Freeze approval

Bind exact result, destination, signer and operation.

03 · STEP

Persist / send

Save intent and signed bytes, then submit locally.

04 · STEP

Verify effect

Read inclusion post-state and matching finalized deploy.

What the example establishes

Verify the approved effect, not merely a successful submission.

Four invariants to preserve

Each statement is bounded by the selected profile, authority model and tested failure model.

I1

Exact-artifact approval

Mutable aliases cannot substitute another result or destination.

I2

Same signed replay

Lost responses do not generate a new timestamp, signature or deployment.

I3

Effect-specific finality

Accepted tuple and successful target deploy effect must match.

I4

Private placement

Episodes, beliefs, provenance and identifying hashes stay off-chain.

Failure is part of the interface

A lost response, a missing prerequisite and a deliberate refusal require different interpretations.

BOUNDARY / TRIGGER	OWNER-AUTHORITATIVE OUTCOME	EVIDENCE / LIMITATION
Caller dies after result acceptance	Recover accepted artifact, reauthorize and continue.	No substitute artifact
Adapter dies before/after sending	Replay persisted signed bytes, even without signer access.	Four total M4 caller/adapter kill boundaries
Node restarts / observation reply lost	Observe same genesis/state; independent reader confirms effect.	No claim of decentralized failover

Recovery evidence covers same-host process termination with intact files; it does not prove power-loss durability.

Measurements and benchmark gaps

Observed values, contract limits and proposed experiments are explicitly different kinds of evidence.

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First finalized-effect observation

One M4 local run; includes proposal/observation overhead

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Persisted signed request

Same bytes survive replay

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Recorded execution cost

Local fixed workflow; not an economic estimate

Single local validator and one generated artifact, with concurrent regressions. No throughput distribution, public-network latency, economic cost or decentralized-safety benchmark is established.

PROPOSED BENCHMARK · NOT RUN

Proposed: repeat the same isolated workload with fixed genesis/config; report inclusion and effect-finality p50/p95 separately, signed size and execution cost, including outage cases.

Current state: what is real, what is not

Current supported regressions pass; historical observations and unsatisfied capabilities keep their labels.

CLASSIFICATION	MODULE-SPECIFIC STATE
REAL / IMPLEMENTED	Pinned F1R3node, actual Hive/DeTTa artifact, post-state query and independent readback.
FIXTURE / LOCAL TEST POLICY	Generated input, test signer/genesis, local approval policy and message counterparty.
BLOCKED / UNSUPPORTED	Exact dirty historical M1 source reproduction remains blocked; clean M4 rebaseline passes separately.
UNRUN	Public networks, multi-validator safety, production keys/economics, power loss and trustless attestation.

M5 final tested runtime: bfb0ec8 · evidence: 3dc6c11 · module representation is not product completeness.

Future directions need acceptance criteria

These are discussion options, not a committed roadmap or permission to expand the current profile.

DIRECTION	PREREQUISITE / DESIGN GATE	MEASURE OF SUCCESS
Measure repeated local behavior	Fixed isolated config and repeated artifact trials	Inclusion and effect-finality distributions separated
Strengthen observation trust	Explicit independent-proof and threat-model contract	Do not confuse node assertions with trustless attestation
Evaluate broader deployment	Separate authorization, isolation and multi-validator study	No silent promotion of local proof into network safety

Keep ownership, current authorization and explicit compatibility intact; qualify new semantics with a new profile.

Evidence you can inspect and reproduce

Read the exact code, contracts and recorded run before treating a claim as a guarantee.

Takeaway

Verify the approved effect, not merely a successful submission.

SUPPORTED REPRODUCTION ENTRYPOINT · PREREQUISITES REQUIRED

```
uv run --python 3.11 --locked python scripts/verify_m4.py <build> --output <fresh>
```

Use the historical tested M4 checkout 55ed540 or its intended compatible scope. M5 verifies unchanged real-chain behavior via verify_m5.py; source, namespace and recipe requirements remain explicit.

`fusion/adapters/f1r3_m4.py`
`integration/hive/commitment/program.py`
`evidence/m4/final/same-artifact.json`
`evidence/m4/final/readback.json`

Source 494df69a1ac401c3d3b2cf79dbfde07f74b5e277 · all citations and full notes accompany this deck.